SCI Soil Health Report Tool

White Paper: Formula Logic, Functions, and Report Calculations

This white paper documents the calculation logic used by the SCI Soil Health Report Tool. The tool converts uploaded field observations, KML field boundaries, and soil lab CSV results into a branded soil health report. It is intended as a transparent methods reference for users reviewing the generated output.

Scope and data inputs

The tool reads a soil-results CSV and, where supplied, a KML/KMZ boundary file. Soil samples are assigned to field boundaries by latitude and longitude point-in-polygon logic. Results are grouped by selected field boundary and by sample depth. Sample depth labels are reported as 0-6 inches / 0-15 cm for surface samples and 6-12 inches / 15-30 cm for sub-surface samples.

Core output sections include summary cards, field boundary map, field observations, Haney results, nutrient cards, aggregate stability, PLFA microbial analysis, notes, and reference range tables. Each section can be toggled on or off before PDF export.

Field observation calculations

Water infiltration

The infiltration fields are interpreted as elapsed minutes for a 1-inch ring infiltration test. The tool preserves the existing over-30-minute logic: readings flagged as greater than 30 minutes are reported as >30 min and are not converted to a numeric rate. Tests 1 and 2 are paired as location A. Tests 3 and 4 are paired as location B.

```
infiltration_rate_in_per_hr = 60 / elapsed_minutes
```

For each sample, the tool calculates rate values for valid numeric readings. A pair average is calculated when paired readings are present; otherwise the average of valid rates is used. The report displays Infiltration A and Infiltration B separately so the two field locations remain visible in the report.

Penetrometer / compaction rating

Spreadsheet penetrometer inputs are treated as inches pushed before the probe reaches the 300 psi maximum. Smaller depth values mean the 300 psi threshold was reached sooner and therefore indicate shallower compaction. A point is counted as compacted when 300 psi is reached within the top 15 inches.

```
percent_points_gt_300psi_top_15in = (points_with_depth_to_300psi <= 15 / total_points) x 100
```

% Points >300 psi in top 15 inches	Compaction rating
<30%	Little-None
30-50%	Slight
50-75%	Moderate
>75%	Severe

The report also displays the average depth at which 300 psi was reached, plus a small bar chart where values at or within the top 15 inches are visually distinguished from deeper readings.

Worm count

Worm counts are reported as counts per cubic foot. The report label is /ft3 or per ft3 depending on the output view.

Slake, VESS, and other field observations

Slake test values are reported as percent of soil remaining after the slake test. VESS score is displayed as entered. Field observations are averaged where multiple samples are grouped into the same field boundary and depth.

Soil lab and report calculations

The tool does not replace laboratory analysis. For lab-supplied values such as Haney soil health calculation, organic matter, pH, nutrients, aggregate stability, and PLFA measures, the report reads the numeric values from the uploaded CSV, groups them by field and depth, and displays the averages or values in formatted report sections.

Common averaging rule

```
For a given field and depth group, numeric values are averaged after blank and non-numeric values are removed.
mean = sum(valid_numeric_values) / count(valid_numeric_values)
```

Reference ratings

Rating bands are used for quick interpretation. These bands are intended for summary screening and should be reviewed in agronomic context, including region, soil texture, crop, management history, and sampling protocol.

Metric	Very Low	Low	Moderate	High	Very High
1:1 Soil pH	<5.5	5.5-6.0	6.0-7.0	7.0-7.5	>7.5
Organic Matter (%)	<2.0	2.0-3.0	3.0-4.0	4.0-5.0	>5.0
CO2-C (ppm)	<10	10-30	30-50	50-100	>100
Organic C:N Ratio	<8	8-10	10-12	12-15	>15
Soil Health Calculation	<5	5-7	7-10	10-15	>15
% MAC	<10	10-20	20-30	30-50	>50

PLFA reference bands

Metric	Very Low	Low	Moderate	High	Very High
Total Microbial Biomass (ng/g)	<1000	1000-1500	1500-2000	2000-2500	>2500
Functional Group Diversity Index	<1.0	1.0-1.1	1.1-1.2	1.2-1.3	>1.3
Total Bacteria (%)	<15	15-20	20-30	30-40	>40
Total Fungi (%)	<1	1-3	3-5	5-10	>10
Fungi:Bacteria Ratio	<0.05	0.05-0.1	0.1-0.2	0.2-0.3	>0.3
Gram+:Gram- Ratio	<0.5	0.5-1.0	1.0-2.0	2.0-4.0	>4

Boundary and sample assignment logic

KML/KMZ polygons are parsed into field boundary and non-field layers. The user can include or exclude detected field boundaries. A soil sample is associated with a boundary when its longitude/latitude point falls inside the selected polygon. Samples outside selected boundaries are omitted from field-boundary rollups unless the user adjusts boundary selections or the source data.

PDF report generation

The report is generated in-browser using the active report options. It uses the same grouped data shown in the on-screen preview. If a section is toggled off, that section is excluded from the PDF output. The map, observation summaries, rating chips, tables, and charts are drawn directly into the PDF export.

Limitations and interpretation notes

The tool is a reporting and screening aid. It depends on the quality of uploaded field/lab data, correct sample coordinates, consistent naming, and sound sampling protocols. Interpretation should be reviewed with local agronomic expertise and should not be treated as a standalone regulatory determination or laboratory certification.